



FORTINET CYBERSECURITY ENGINEERING TRACK

FINAL PROJECT

IMPLEMENTING VPN SOLUTIONS WITH FORTIGATE

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Introduction

Virtual Private Networks (VPNs) are fundamental technologies that enable secure communication over untrusted networks. As modern organizations increasingly depend on distributed workforces, cloud services, and multi-site infrastructures, VPNs provide confidentiality, integrity, and reliable remote access to corporate resources. They protect traffic using encryption and tunneling mechanisms, allowing users and branches to exchange data securely as if they were on the same private network.

Fortinet's FortiGate platform offers a robust suite of VPN solutions that combine security, performance, and ease of deployment. Beyond basic connectivity, FortiGate integrates advanced capabilities such as threat inspection, identity-based access, and intelligent routing through SD-WAN. These features make it a powerful option for enterprises requiring both secure remote access and scalable site-to-site connectivity.

This project explores three major VPN implementations using FortiGate. In the first stage, an SSL VPN is configured to enable secure remote user access. The second stage establishes an IPsec site-to-site tunnel between two FortiGate devices to support branch interconnectivity. Finally, the project integrates FortiGate SD-WAN to optimize VPN traffic by selecting the best available WAN link for performance and reliability. The results of these implementations are documented, tested, and analyzed to demonstrate how Fortinet technologies can deliver efficient and secure VPN solutions.

Project Objectives

The main objective of this project is to design, configure, and evaluate secure VPN solutions using FortiGate. The project aims to explore how different VPN technologies address real-world requirements, such as secure remote access, site-to-site communication, and traffic optimization across multiple WAN links.

To achieve this, the project focuses on three key implementation stages:

- 1. Understanding VPN Concepts and SSL VPN Setup**

Study different types of VPNs and their use cases, then configure an SSL VPN on FortiGate to enable secure remote user access.

- 2. Implementing IPsec VPN Connectivity**

Establish a secure IPsec tunnel between two FortiGate devices, enabling encrypted site-to-site communication between private networks.

- 3. Integrating SD-WAN with VPN Solutions**

Configure SD-WAN features on FortiGate to monitor WAN links and intelligently route VPN traffic to improve reliability and performance.

Through these tasks, the project aims to demonstrate practical VPN deployment steps, compare different VPN methods, and illustrate how SD-WAN enhances VPN traffic stability in enterprise environments.

Theoretical Background

1. Virtual Private Networks (VPNs)

A **Virtual Private Network (VPN)** is a secure communication method that creates an encrypted tunnel between private or public networks. VPNs allow confidential data to travel across untrusted infrastructure—such as the internet—while maintaining confidentiality, integrity, and authentication.

Key Security Principles

- **Confidentiality** – Preventing unauthorized access to transmitted data through encryption.
- **Integrity** – Ensuring data is not modified during transit.
- **Authentication** – Verifying the identity of endpoints or users.

VPNs are widely used in enterprise environments to:

- Enable remote employee access to internal resources
- Connect branch offices securely (site-to-site)
- Improve cloud, IoT, or data center connectivity

2. SSL VPN

Concept

SSL VPN (Secure Sockets Layer Virtual Private Network) uses HTTPS (TLS) to secure communication between a user and the organization's internal resources. Unlike IPsec tunneling, SSL VPN works at the application layer (Layer 7) and is delivered through a web browser or VPN client.

Characteristics

- **Remote-access focused** — best for employees connecting from outside the network
- **Port-friendly** — runs over TCP/443, bypassing network restrictions

- **User authentication** — credentials, certificates, or MFA
- **Client-based or clientless** — depending on the deployment

Use Cases

- Secure employees accessing internal portals
- Remote management of servers or applications
- BYOD scenarios (phones, tablets, laptops)

In this project, the SSL VPN provides easy remote access to network resources through FortiGate, forming the foundation for VPN learning and validation.

3. IPsec VPN

Concept

IPsec (Internet Protocol Security) is a framework of protocols that secures IP communication at the network layer (Layer 3). It is the most common standard for site-to-site or branch-to-branch encryption.

Core Components

1. **IKE (Internet Key Exchange)** – Negotiates cryptographic parameters and authentication.
2. **ESP (Encapsulating Security Payload)** – Encrypts and protects payload data.
3. **Tunnel Mode** – Wraps original IP packets inside encrypted IPsec packets.

Advantages

- Strong encryption (AES, 3DES)
- High throughput for branch networks
- Interoperability between vendors
- Suitable for static or permanent tunnels

Use Cases

- Linking corporate headquarters to branch offices
- Secure inter-data center communication
- IoT/M2M connections

In this project, an IPsec tunnel is established between two FortiGate devices, simulating branch connectivity and validating tunnel integrity through connectivity testing.

4. SD-WAN (Software-Defined Wide Area Network)

Concept

SD-WAN is a network management approach that dynamically monitors WAN link performance and routes traffic based on real-time conditions. It allows enterprises to use multiple ISP links simultaneously and apply intelligent routing rules to maximize performance.

How SD-WAN Enhances VPNs

- Monitors latency, jitter, and packet loss
- Automatically selects the optimal WAN path
- Distributes overlays (e.g., IPsec) across multiple WAN links
- Provides failover during link degradation or failure

Key Benefits

- Improved end-to-end VPN performance
- Application-aware routing (VoIP, streaming, corporate apps)
- Higher availability through link redundancy
- Reduced operational cost vs MPLS

In this project, SD-WAN is configured on FortiGate to optimize VPN traffic, demonstrating how modern enterprises improve service quality without compromising security.

Implementation and Results

FortiGate SSL VPN Configuration

1. Network Design (Week 1)

The lab environment for Week 1 uses the following setup:

Laptop → Home Router → VMware → FortiGate VM

- **WAN (port1):** Receives DHCP IP from home router
- **LAN (port2):** Static IP 192.168.100.1/24
- **SSL-VPN Address Pool:** 10.212.134.200 – 10.212.134.210
- **Access:** FortiGate Web GUI

2. Week 1 Scope

The following tasks are included in Week 1:

- Configure WAN and LAN interfaces
- Create Local User for VPN authentication
- Create User Group for SSL VPN users
- Configure SSL-VPN Settings
- Configure SSL-VPN Portal (Tunnel Mode)
- Assign Tunnel Address Pool
- Create Firewall Policies for VPN traffic
- Document all steps with screenshots

3. Summary

Week 1 focuses on understanding VPN concepts and implementing the complete SSL VPN remote access configuration on FortiGate.

All components (interfaces, users, portal, address pool, and policies) are prepared for testing in the upcoming project weeks.

4. Configuration Steps

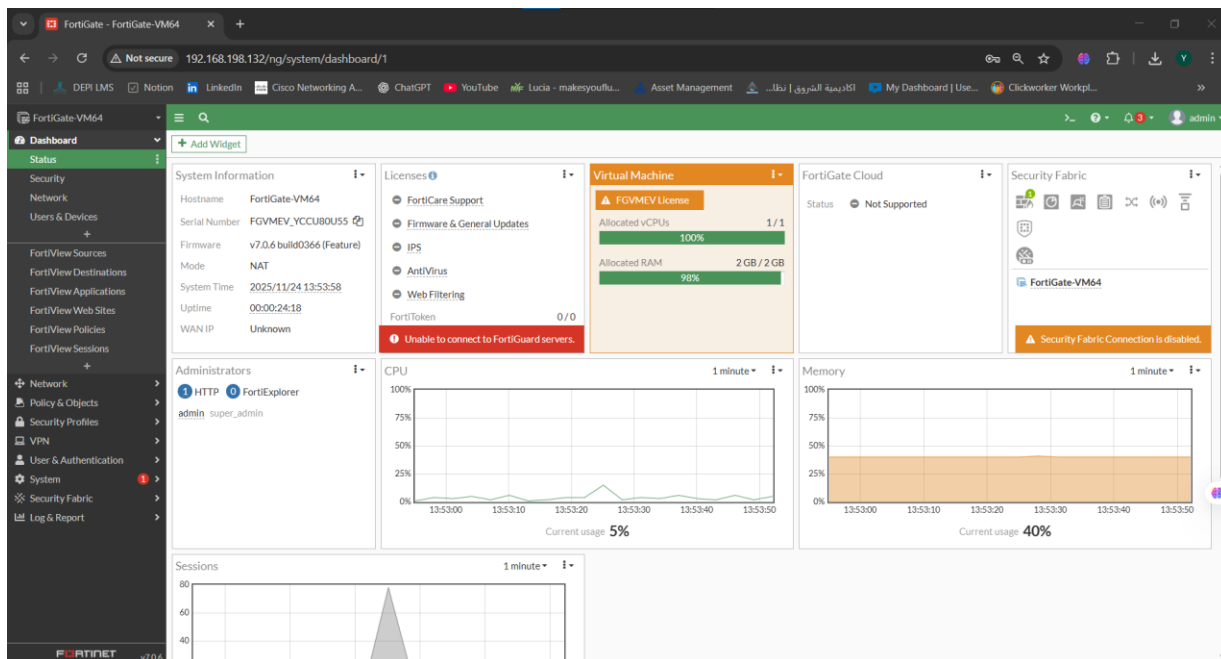


Figure 1 - Shows the main system summary, firmware version, uptime, and status of all interfaces.

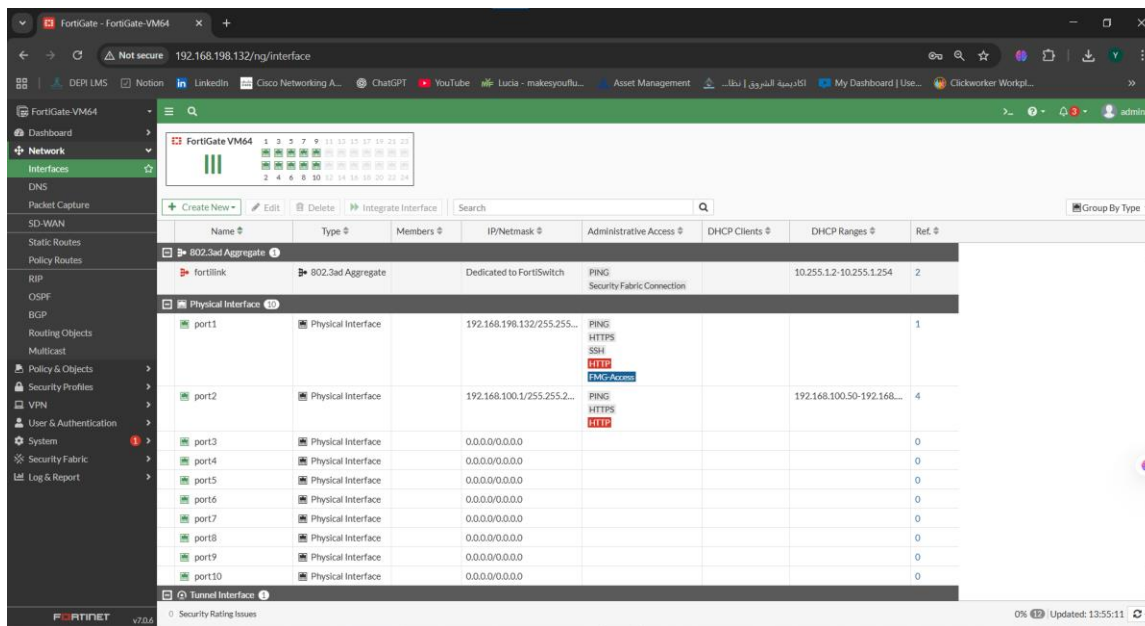


Figure 2 – (Network Interfaces Overview) Displays all FortiGate interfaces including port1 (WAN) and port2 (LAN).

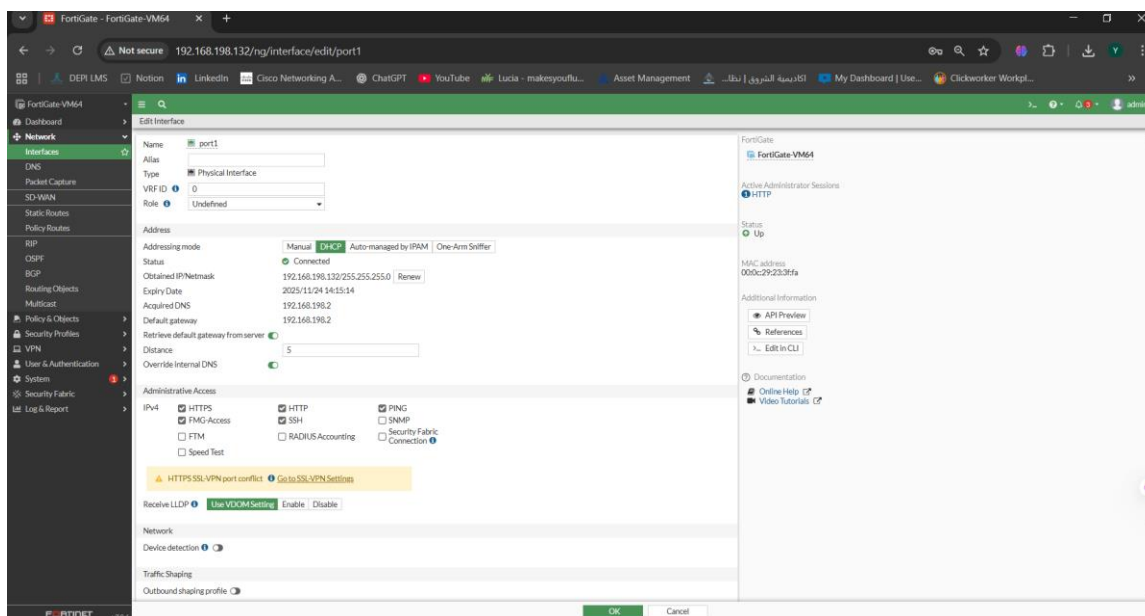


Figure 3 – (Port1 (WAN) Configuration) Port1 configured as the WAN interface using DHCP with assigned IP address.

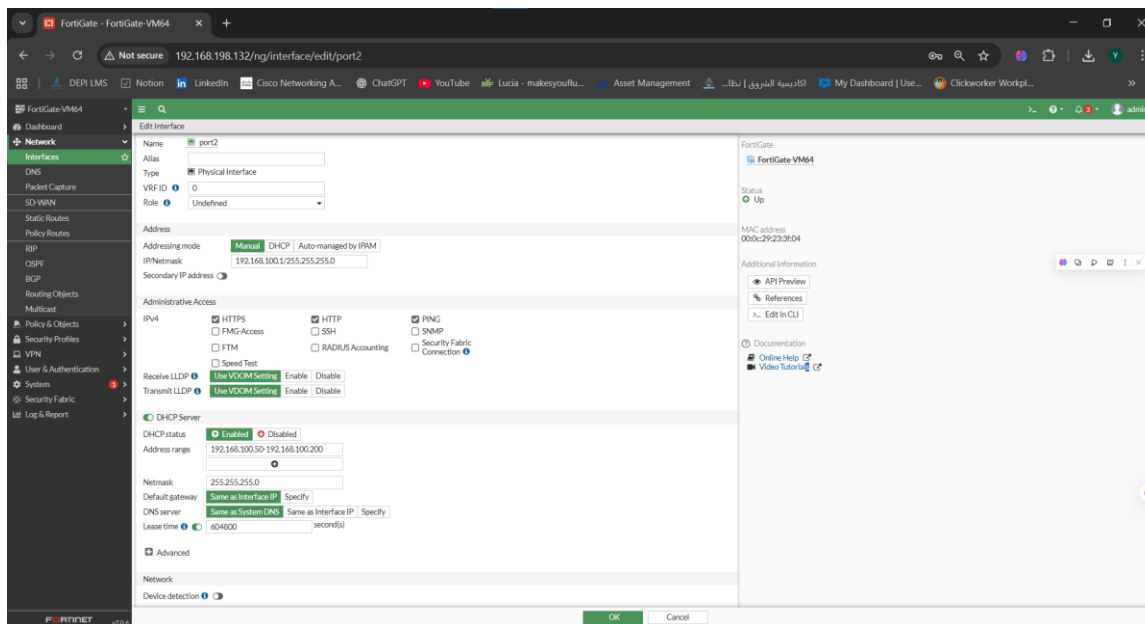


Figure 4 – (Port2 (LAN) Configuration) Port2 configured as the internal LAN interface with IP 192.168.100.1/24

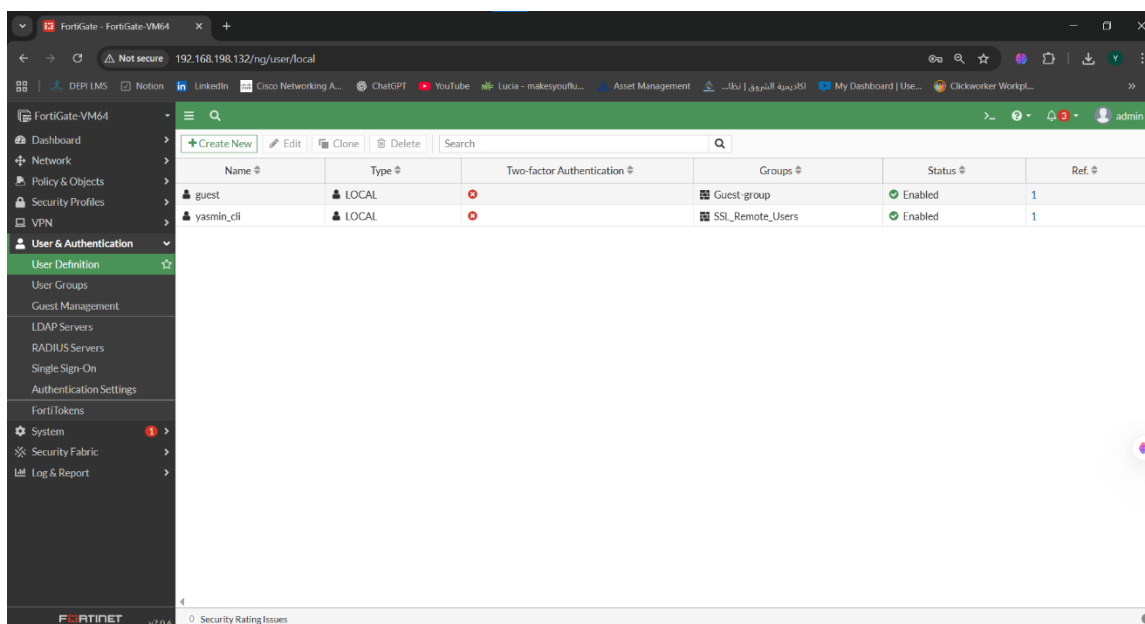


Figure 5 – (Local User (yasmin_cli)) Shows creation of the local user used for SSL-VPN authentication

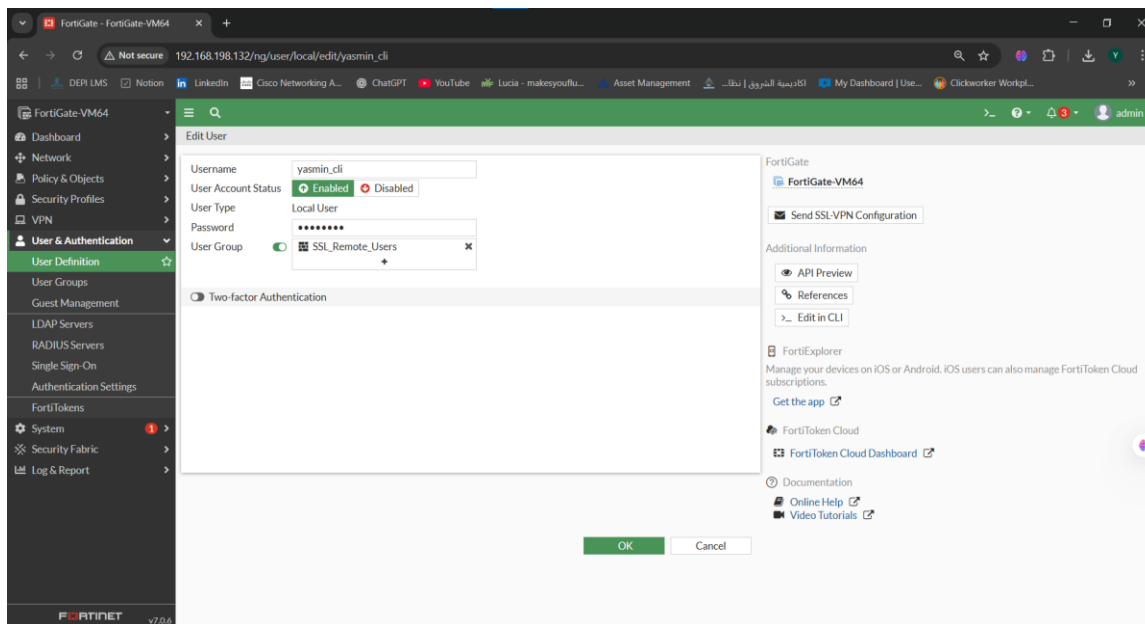


Figure 6 – (Local User Details) Displays full configuration of the user yasmin_cli, including password and status

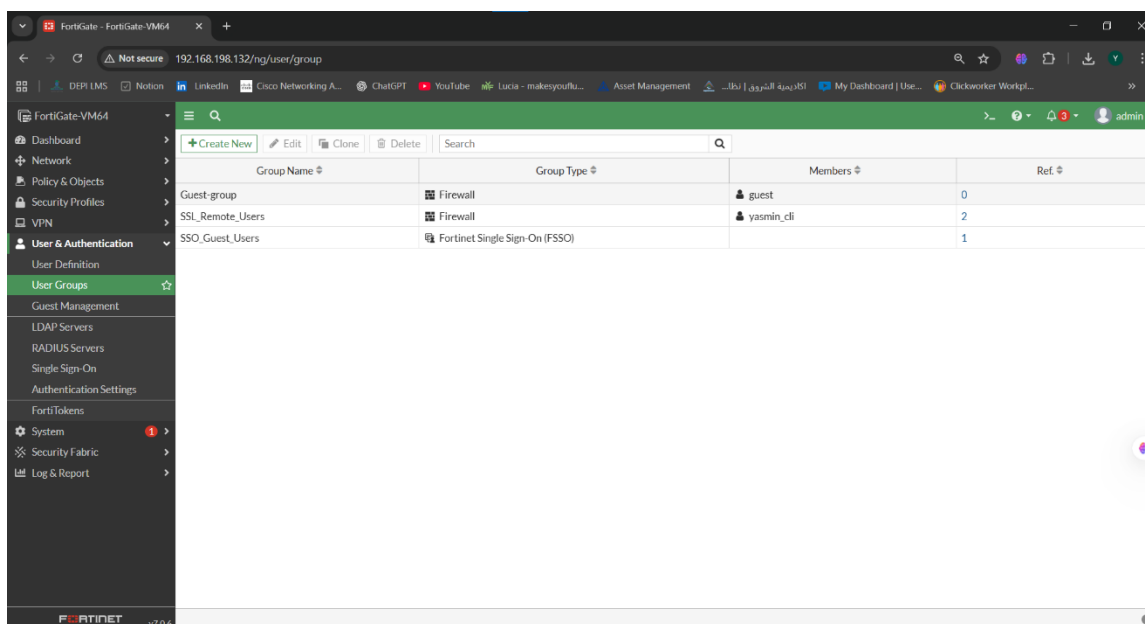


Figure 7 – (SSL_Remote_Users) Group User group created to manage SSL-VPN remote access user

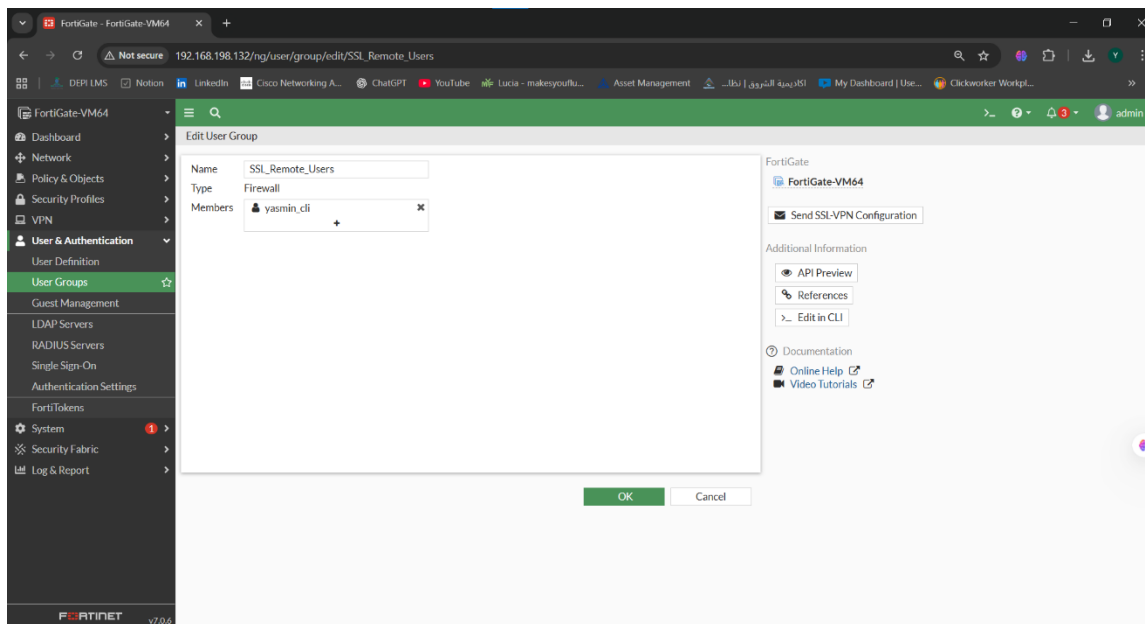


Figure 8 – (Group Members) Shows the user yasmin_cli added to the SSL_Remote_Users group

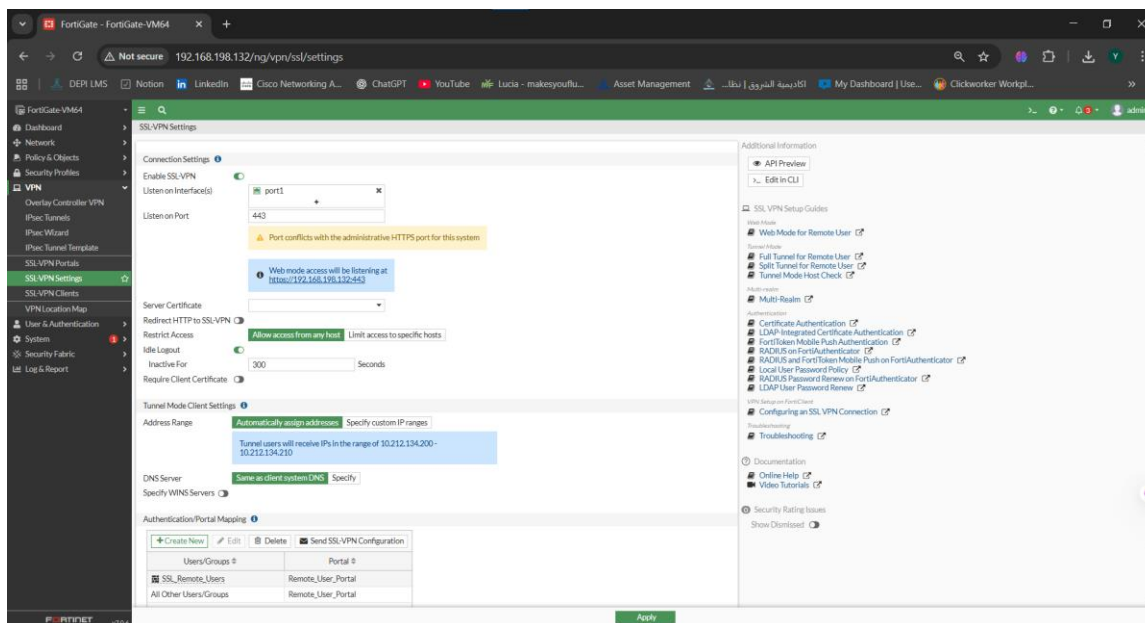


Figure 9 – (SSL-VPN Settings (Main Page)) SSL-VPN enabled on port1 with tunnel mode and address pool configuration.

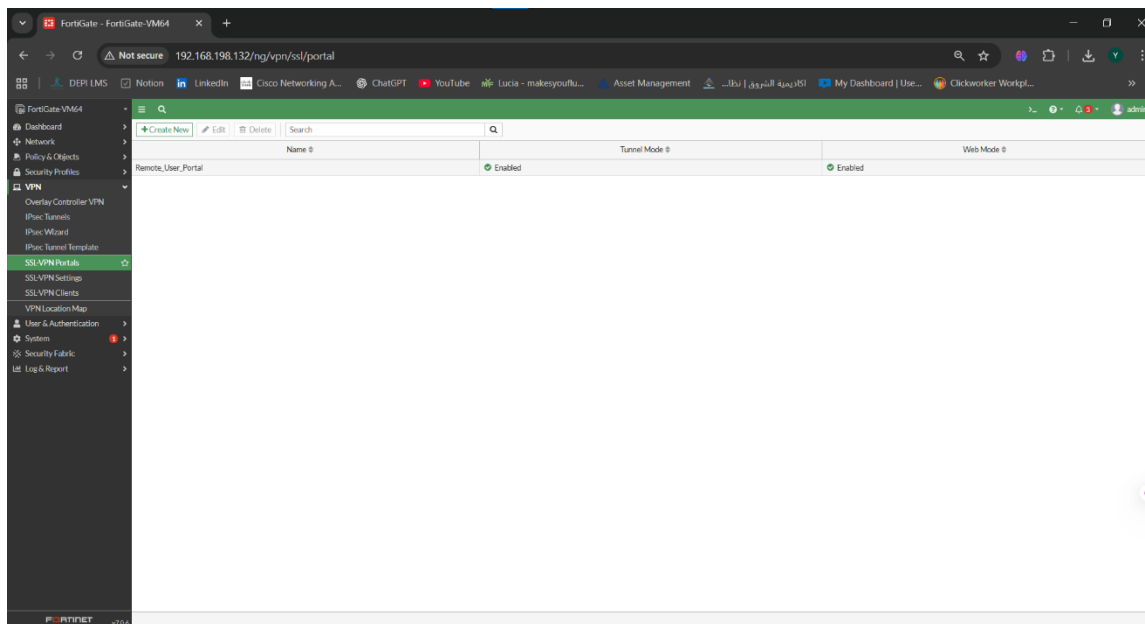


Figure 10 – (SSL-VPN Portal (Remote_User_Portal)) Portal settings for remote users including split tunneling and routing options.

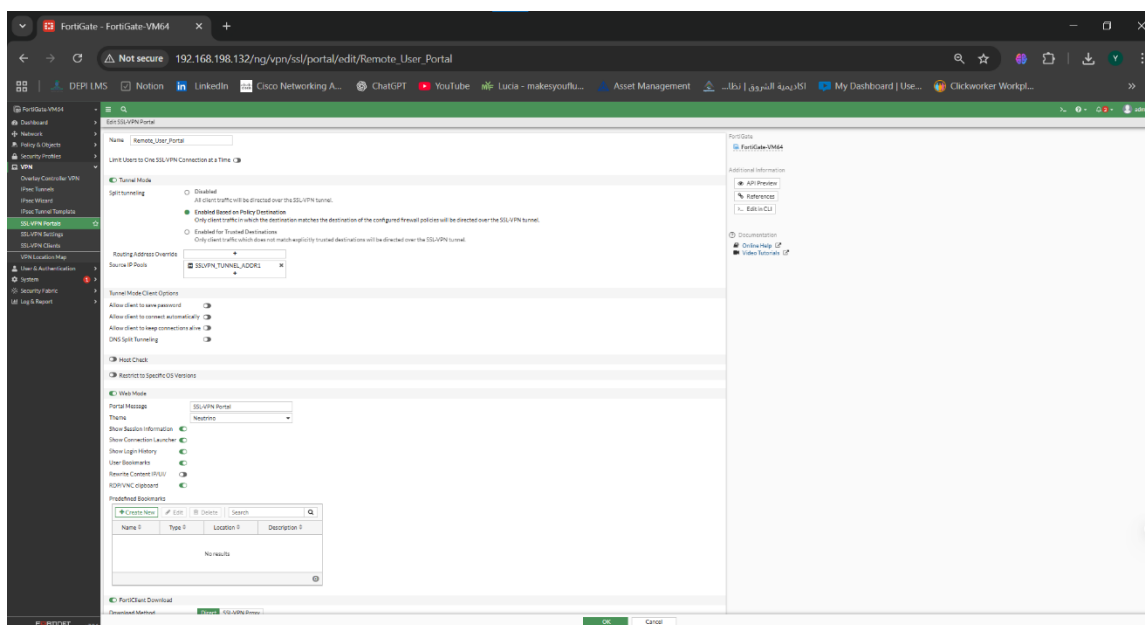


Figure 11 - SSL-VPN Portal Configuration (Remote_User_Portal) with Split-Tunneling Enabled and Routing Address Override Set to SSLVPN_TUNNEL_ADDR1

Figure 11 → This configuration defines the SSL-VPN user portal settings, including tunnel mode behavior, split tunneling based on policy destination, and the IP pool assigned to remote users

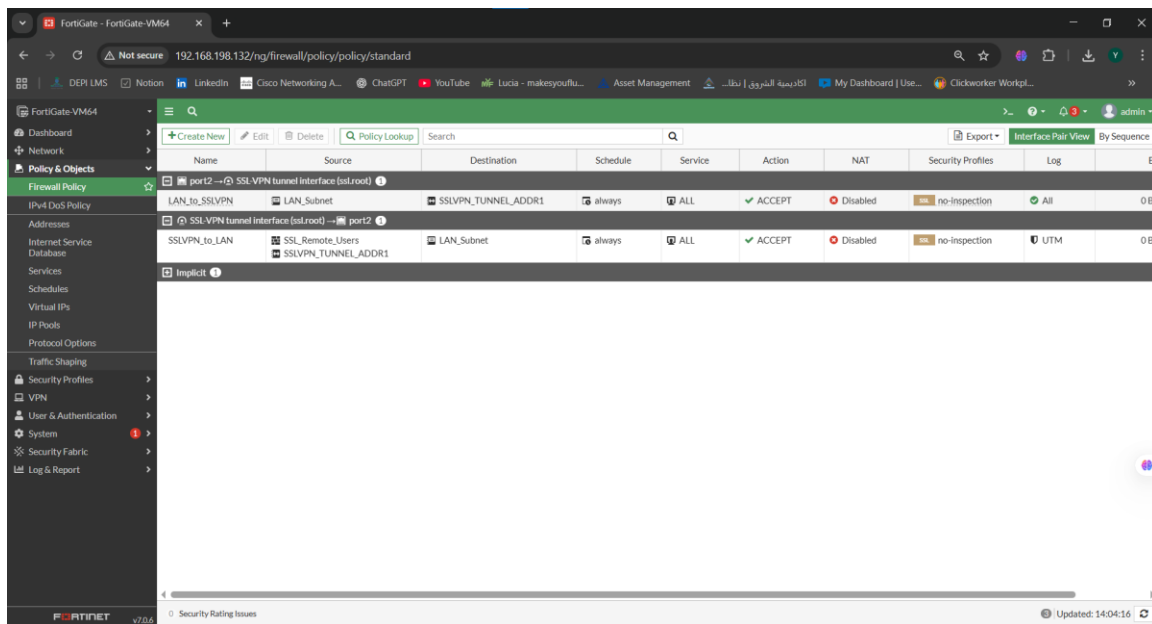


Figure 12 – (Firewall Policies Overview) Overview of the SSL-VPN firewall policies, including LAN_to_SSLVPN and SSLVPN_to_LAN rules.

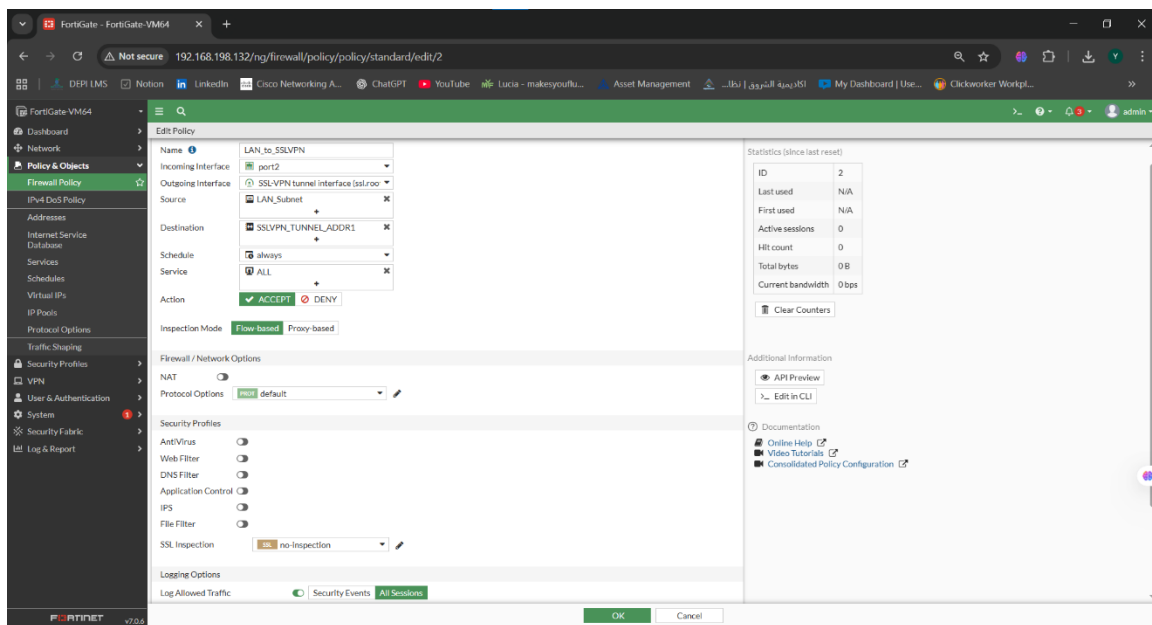


Figure 13 – (Firewall Policy: LAN → SSLVPN) firewall Policy allowing LAN_Subnet to communicate with SSLVPN_TUNNEL_ADDR1 through port2 → ssl.root

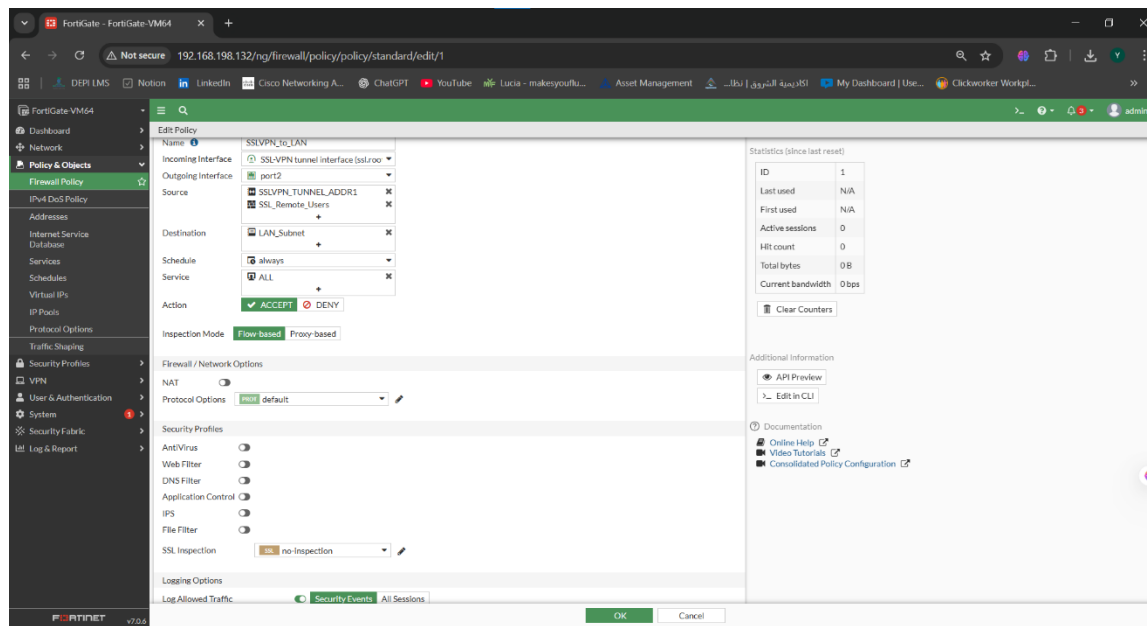


Figure 14 – (Firewall Policy) SSLVPN → LAN Policy allowing SSL-VPN users to access the LAN subnet

4. Test Results

Due to the limitations of the FortiGate VM (no SSL-VPN license),

VPN connection testing could not be performed in Week 1.

However, all required SSL-VPN configuration steps were completed successfully, including:

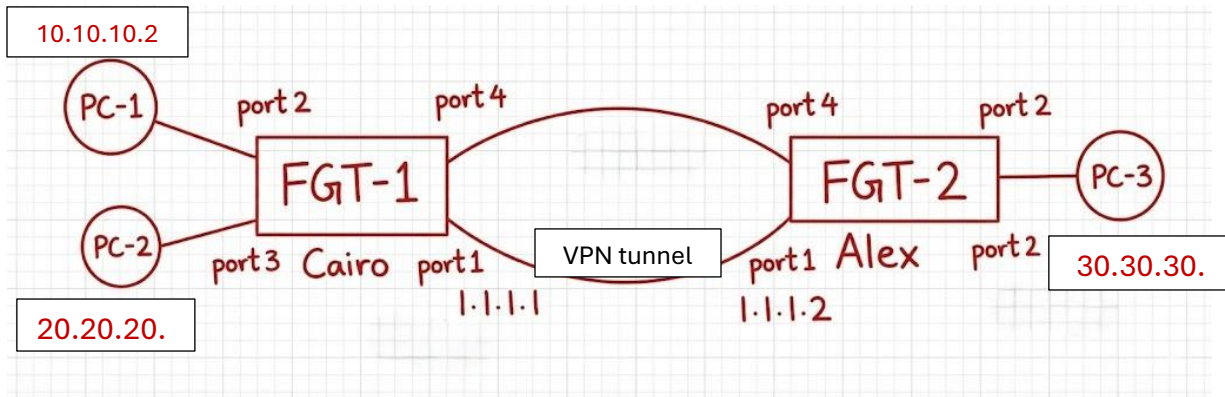
- Interface configuration
- Local user creation
- User group assignment
- SSL-VPN portal setup
- Tunnel IP range configuration
- Firewall policies for SSL-VPN traffic

5. Conclusion

All SSL-VPN configurations required for Week 1 have been successfully completed, including interfaces, users, groups, SSL-VPN settings, portals, and firewall policies. Connection testing is not required in this phase.

IPsec VPN Configuration

We established this topology:



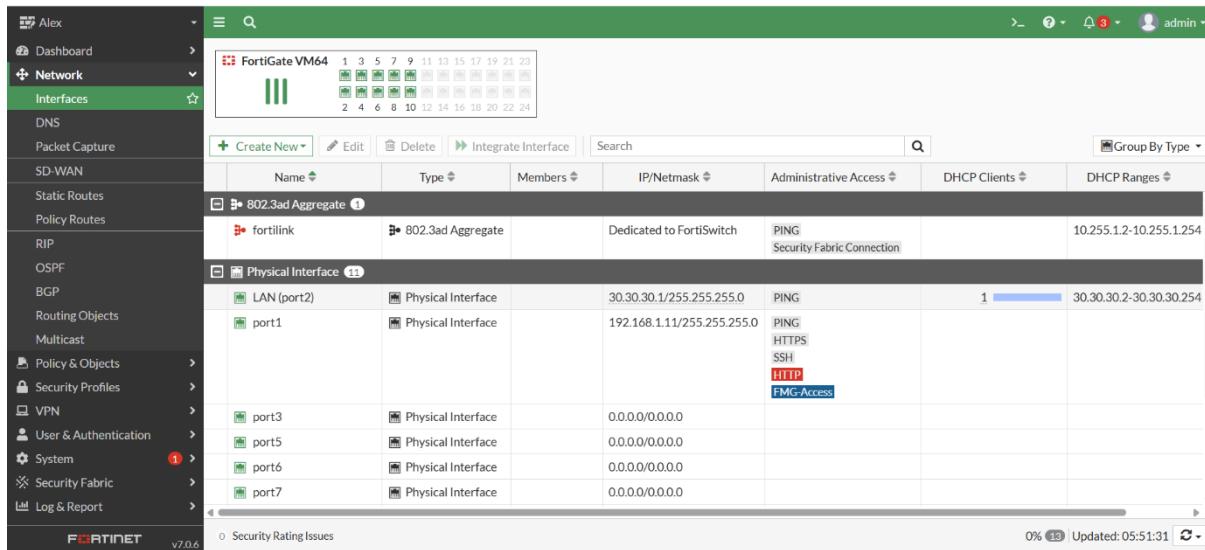
1.Configuration:

1. Configure the interfaces and make port 1 is the management port

Cairo:

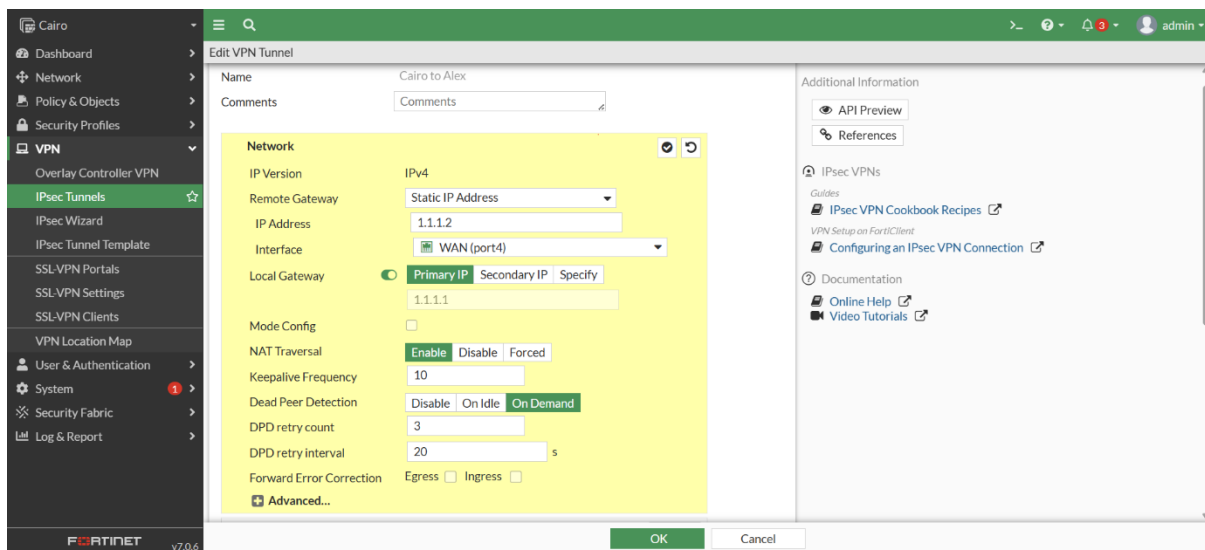
Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254
LAN 1 (port2)	Physical Interface		10.10.10.1/255.255.255.0	PING	1	10.10.10.2-10.10.10.254
LAN 2 (port3)	Physical Interface		20.20.20.1/255.255.255.0	PING	1	20.20.20.2-20.20.20.254
port1	Physical Interface		192.168.1.13/255.255.255.0	PING HTTPS SSH HTTP FIMG Access		
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			

Alex:



2. Configure the vpn tunnel between the two firewalls and choose traffic selector from the network 10.10.10.0 to 30.30.30.0 in Cairo firewall and reverse in the Alex firewall

Cairo:



Cairo

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

Edit VPN Tunnel

Remote Gateway : Static IP Address (1.1.1.2) , Local Gateway: 1.1.1.1, Interface : port4

Authentication

MethodPre-shared Key

Pre-shared Key123456

IKE

Version12

ModeAggressiveMain (ID protection)

Phase 1 Proposal

+

Add

EncryptionDES

AuthenticationMD5

32

31

30

29

28

27

21

20

19

18

17

16

15

14

5

2

1

Diffie-Hellman Groups

Key Lifetime (seconds)

86400

Local ID

XAUTH

TypeDisabled

Cairo

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

Edit VPN Tunnel

Type : Disabled

Phase 2 Selectors

Name	Local Address	Remote Address
10 - 30	10.10.10.0/255.255.255.0	30.30.30.0/255.255.255.0

Edit Phase 2

Name

10 - 30

Comments

Comments

Local Address

Subnet

10.10.10.0/255.255.255

Remote Address

Subnet

30.30.30.0/255.255.255

Advanced...

Phase 2 Proposal

Add

Encryption

DES

Authentication

MD5

Encryption

DES

Authentication

SHA1

Enable Replay Detection

Enable Perfect Forward Secrecy (PFS)

Local Port

All

Remote Port

All

Protocol

All

Auto-negotiate

Autokey Keep Alive

Key Lifetime

Seconds

Seconds

43200

OK

Cancel

Cairo

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

Create New

Edit

Delete

Search

Tunnel	Interface Binding	Status	Ref.
Cairo to Alex	WAN (port4)	Up	4

Security Rating Issues

Alex:

Edit VPN Tunnel

Name: Alex to Cairo

Comments: Comments

Network

IP Version: IPv4

Remote Gateway: Static IP Address

IP Address: 1.1.1.1

Interface: WAN (port4)

Local Gateway: ☒ Primary IP: 1.1.1.2 ☐ Secondary IP: Specify

Mode Config: ☐

NAT Traversal: ☒ Enable ☐ Disable ☐ Forced

Keepalive Frequency: 10

Dead Peer Detection: ☒ Disable ☐ On Idle ☐ On Demand

DPD retry count: 3

DPD retry interval: 20 s

Forward Error Correction: Egress ☐ Ingress ☐

Advanced...

Additional Information

API Preview

References

IPsec VPNs

Guides

- IPsec VPN Cookbook Recipes
- VPN Setup on FortiClient
- Configuring an IPsec VPN Connection

Documentation

- Online Help
- Video Tutorials

Edit VPN Tunnel

Name: Alex to Cairo

Comments: Comments

Network

Remote Gateway: Static IP Address (1.1.1.1), Local Gateway: 1.1.1.2, Interface: port4

Authentication

Method: Pre-shared Key

Pre-shared Key: 123456

IKE

Version: ☒ 1 ☐ 2

Mode: ☒ Aggressive ☐ Main (ID protection)

Phase 1 Proposal

Algorithms: DES-MD5

Diffie-Hellman Groups: 14, 5, 2

XAUTH

Type: Disabled

Additional Information

API Preview

References

IPsec VPNs

Guides

- IPsec VPN Cookbook Recipes
- VPN Setup on FortiClient
- Configuring an IPsec VPN Connection

Documentation

- Online Help
- Video Tutorials

Phase 1 Proposal

Encryption: DES

Authentication: MD5

Diffie-Hellman Groups: ☐ 32 ☐ 31 ☐ 30 ☐ 29 ☐ 28 ☐ 27 ☐ 21 ☐ 20 ☐ 19 ☐ 18 ☐ 17 ☐ 16 ☒ 15 ☒ 14 ☒ 5 ☒ 2 ☐ 1

Key Lifetime (seconds): 86400

Local ID:

XAUTH

Type: Disabled

Alex

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

Edit VPN Tunnel

Edit Phase 2

Name30 - 10

Comments

Local AddressSubnet30.30.30.0/255.255.255

Remote AddressSubnet10.10.10.0/255.255.255

Advanced...

Phase 2 ProposalAdd

EncryptionDESAuthenticationMD5

EncryptionDESAuthenticationSHA1

Enable Replay Detection

Enable Perfect Forward Secrecy (PFS)

Local PortAll

Remote PortAll

ProtocolAll

Auto-negotiate

Autokey Keep Alive

Key LifetimeSeconds

Seconds43200

Alex

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

Create NewEditDeleteSearch

Tunnel	Interface Binding	Status	Ref.
Alex to Cairo	WAN (port4)	Up	4

0 Security Rating Issues

3.Configure the static route

Cairo:

Cairo

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

≡

Q

+ Create New

Edit

Clone

Delete

Search

Q

Destination	Gateway IP	Interface	Status	Comments
30.30.30.0/24	1.1.1.2	Cairo to Alex	Enabled	

Alex:

Alex

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FORTINET

v7.0.6

≡

Q

+ Create New

Edit

Clone

Delete

Search

Q

Destination

Gateway IP

Interface

Status

Comments

10.10.10.0/24

1.1.1.1

Alex to Cairo

Enabled

4. Configure the policy between network 10.10.10.0 and 30.30.30.0

Cairo:

The screenshot shows the Fortinet Cairo Firewall Policy configuration page. The left sidebar contains a navigation menu with options like Dashboard, Network, Policy & Objects, and various security features. The main area displays a table of firewall policies. The 'Firewall Policy' section is active, showing a list of policies. The 'Policy Lookup' tab is selected, displaying a table with columns: Name, Source, Destination, Schedule, Service, Action, NAT, Security Profiles, Log, and Bytes. The table lists three policies: 'Cairo to Alex -> LAN 1 (port2)', 'LAN 1 (port2) -> Cairo to Alex', and 'LAN1 TO VPN'. The 'Implicit' policy is also listed at the bottom.

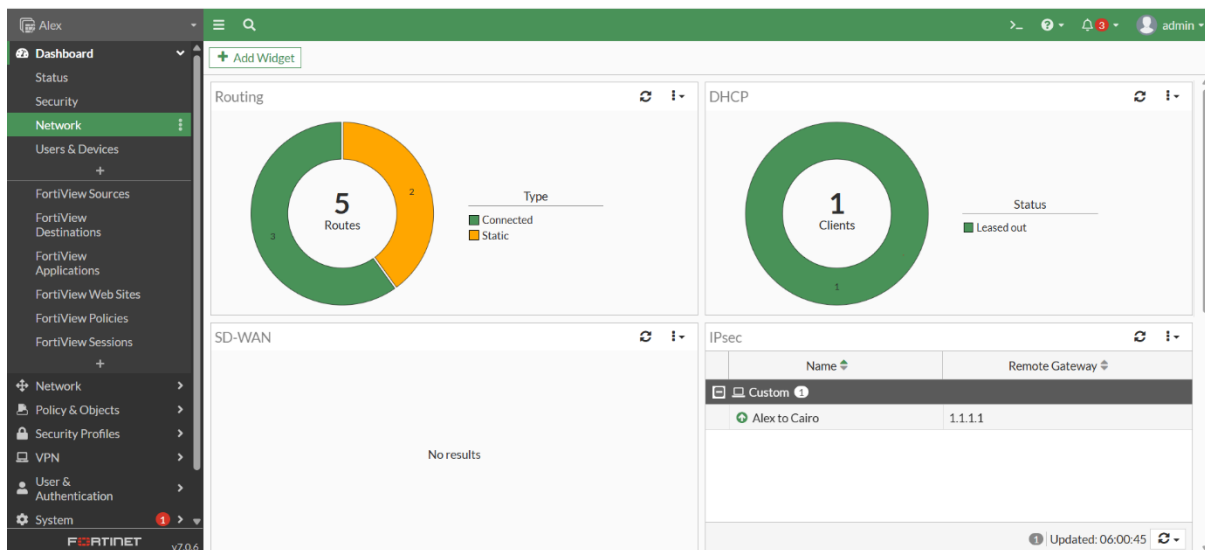
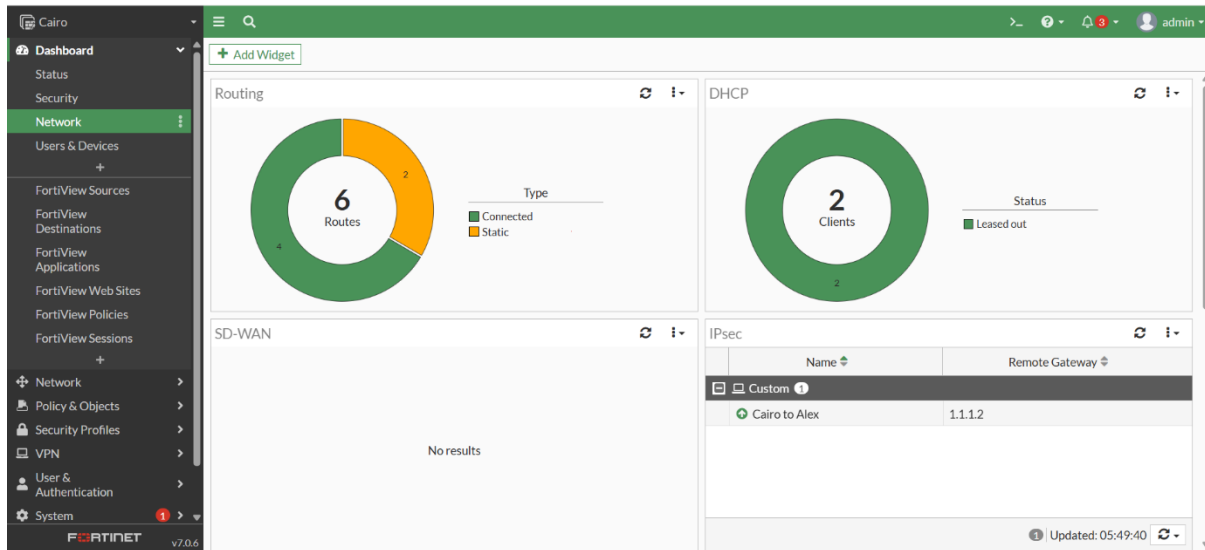
Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	
Cairo to Alex -> LAN 1 (port2)	VPN TO LAN1	NET 30	NET 10	always	ALL	ACCEPT	Disabled	no-inspection	All	547.34 kB
LAN 1 (port2) -> Cairo to Alex	LAN 1 (port2)	Cairo to Alex								
LAN1 TO VPN	NET 10	NET 30	always	ALL	ACCEPT	Disabled	no-inspection	All		54.26 kB
Implicit										

Alex:

The screenshot shows the Fortinet Alex Firewall Policy configuration page. The left sidebar contains a navigation menu with options like Dashboard, Network, Policy & Objects, and various security features. The main area displays a table of firewall policies. The 'Policy Lookup' tab is selected, displaying a table with columns: Name, Source, Destination, Schedule, Service, Action, NAT, Security Profiles, Log, and Bytes. The table lists three policies: 'Alex to Cairo -> LAN (port2)', 'LAN (port2) -> Alex to Cairo', and 'LAN.VPN'. The 'Implicit' policy is also listed at the bottom.

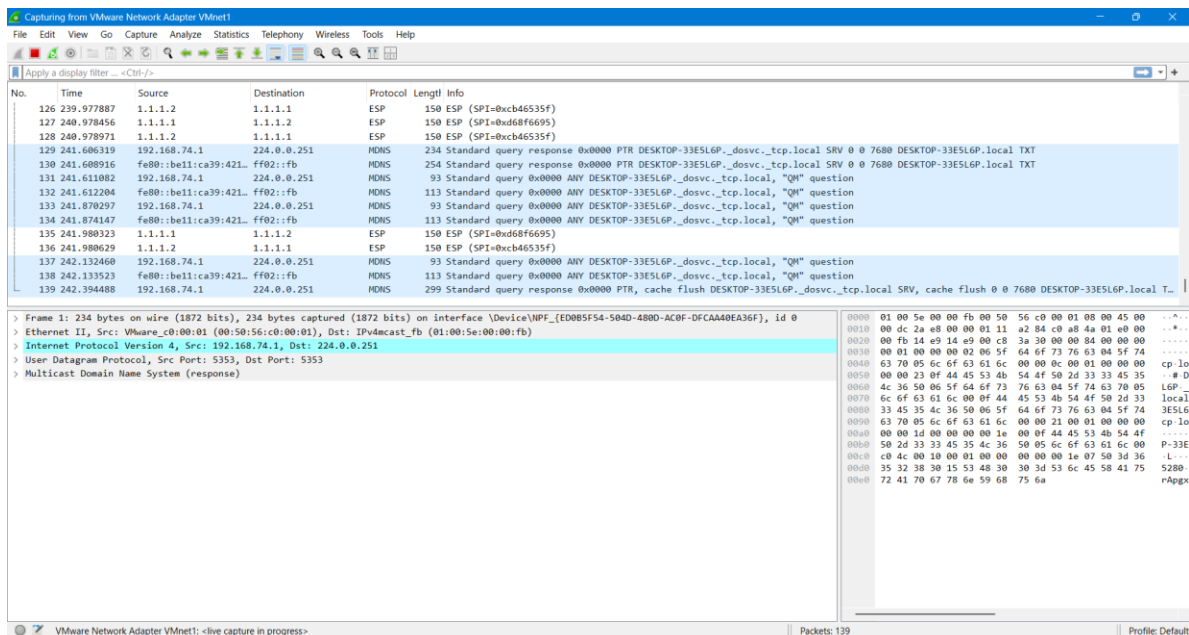
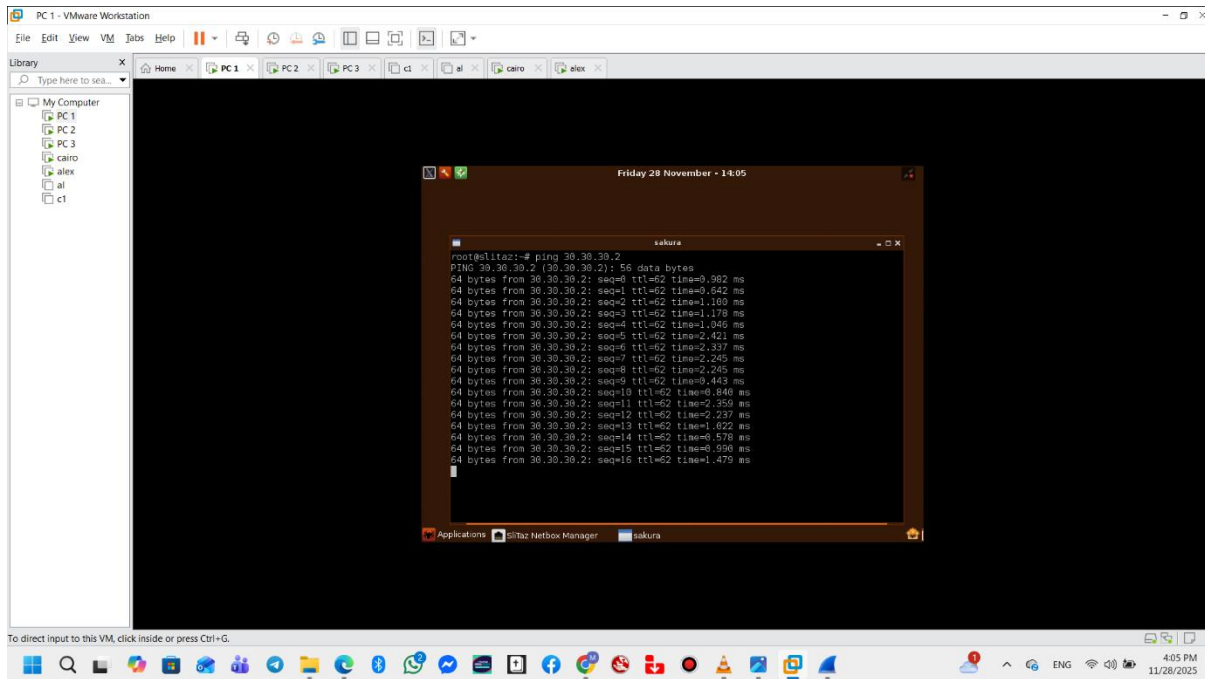
Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	
Alex to Cairo -> LAN (port2)	VPN-LAN	NET 10	NET 30	always	ALL	ACCEPT	Disabled	no-inspection	All	0 B
LAN (port2) -> Alex to Cairo	LAN (port2)	Alex to Cairo								
LAN.VPN	NET 30	NET 10	always	ALL	ACCEPT	Disabled	no-inspection	All		0 B
Implicit										

So the Vpn tunnel now is up and running

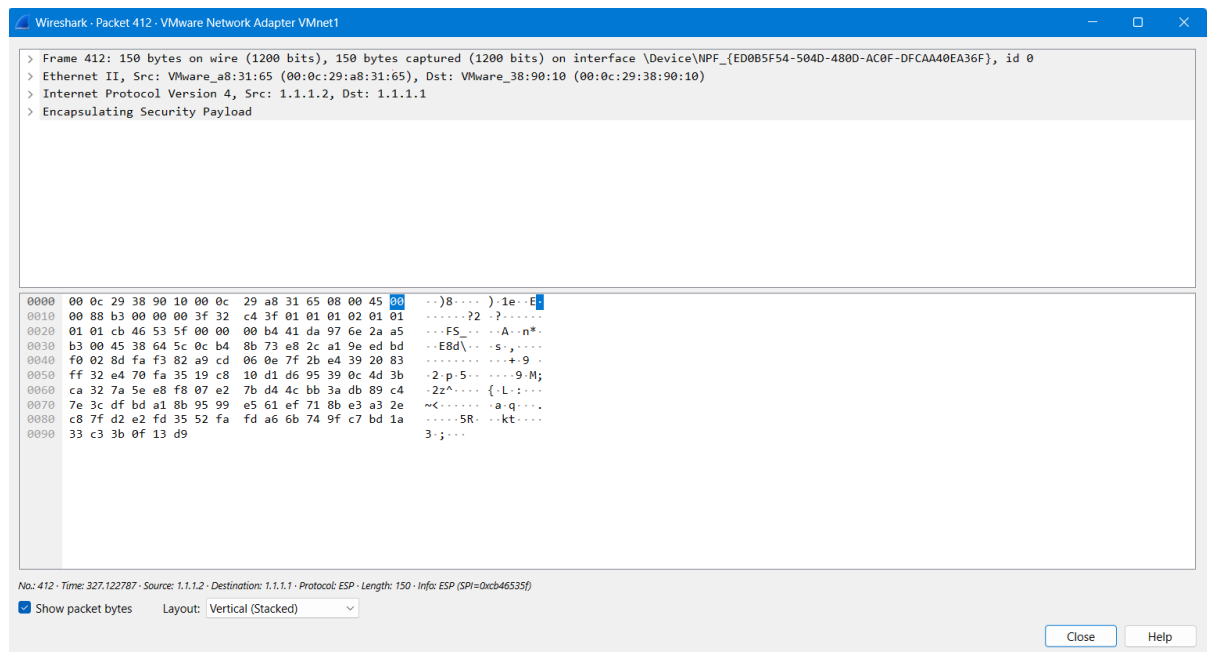
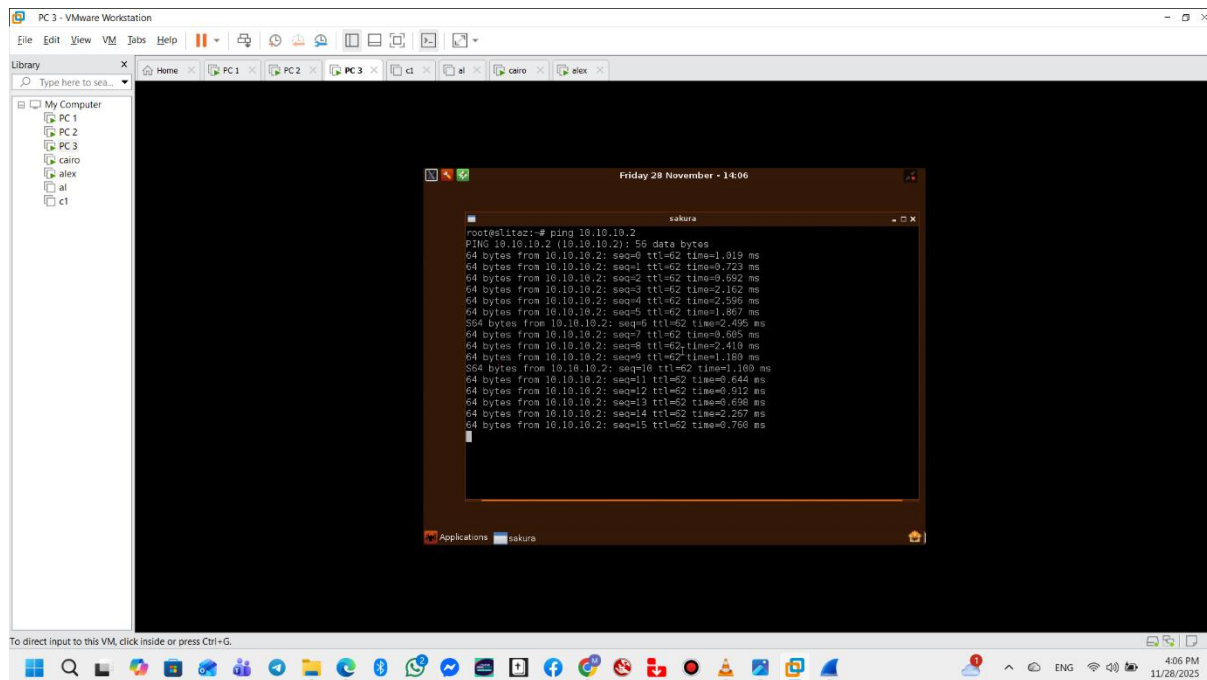


Test Results

So let us see the result when we ping from PC-1 to PC-3



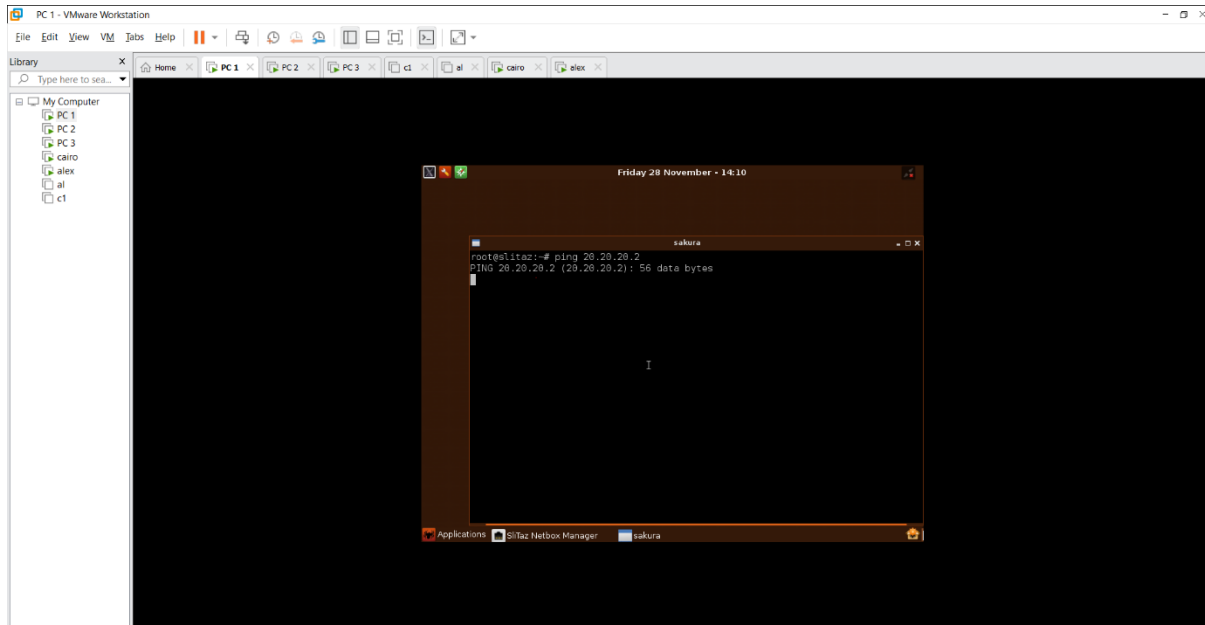
From PC-3 to PC-1



Comment:

The connectivity test results between PC-1 and PC-3 are successful because we established traffic selector from the network 10.10.10.0 to 30.30.30.0

Ping from PC-1 to PC-2



Comment:

The connectivity test results between PC-1 and PC-2 have failed because there is no traffic selector between network 10 and network 20 so the result is as expected

Fortigate VPN with SD-WAN

Configuration of SD-WAN on fortigate

1. Configure the ports to a wan

The screenshot displays the FortiGate configuration interface for a device named 'Cairo'. The left sidebar shows the navigation menu with 'Network' and 'Interfaces' selected. The main area shows the 'Edit Interface' configuration for 'wan 2 (port3)'. The configuration includes fields for Name, Alias, Type, VRF ID, Role, Estimated bandwidth, Addressing mode, Status, Obtained IP/Netmask, Expiry Date, Acquired DNS, Default gateway, Retrieve default gateway from server, Distance, Override internal DNS, and Administrative Access. The 'Status' is 'Connected' and the 'Administrative Access' is 'PING'. A red banner at the top indicates 'FortiGate time is out of sync.'.

Below the configuration window, the 'Interfaces' table is displayed, showing a list of interfaces and their configurations. The table includes columns for Name, Type, Members, IP/Netmask, Administrative Access, DHCP Clients, and DHCP Ranges.

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			
port8	Physical Interface		0.0.0.0/0.0.0.0			
port9	Physical Interface		0.0.0.0/0.0.0.0			
port10	Physical Interface		0.0.0.0/0.0.0.0			
WAN (port4)	Physical Interface		1.1.1.1/255.255.255.0	PING		
wan 2 (port3)	Physical Interface		30.30.30.2/255.255.255.0	PING		
SD-WAN Zone						
virtual-wan-link	SD-WAN Zone	WAN (port4) wan 2 (port3)	0.0.0.0/0.0.0.0			
Tunnel Interface						
NAT interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0			

2. Create new SD-WAN member

Cairo

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate time is out of sync.

Edit SD-WAN Member

Interface

SD-WAN Zone

Gateway

Cost

Priority

Status

None

virtual-wan-link

0.0.0.0

0

1

Enabled

API Preview

SD-WAN Setup Guides

Creating the SD-WAN Interface

MPLS (SIP and Backup) + DIA (Cloud Apps)

SD-WAN Traffic Shaping and QoS with SD-WAN

Per Packet Distribution and Tunnel Aggregation

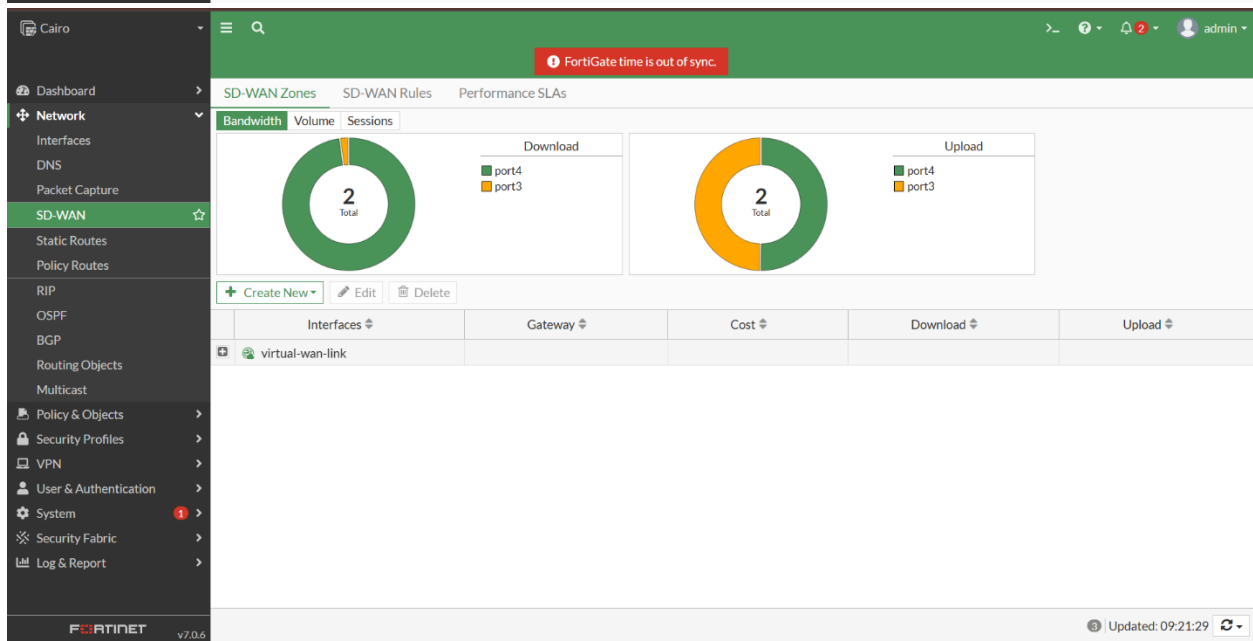
Documentation

Online Help

Video Tutorials

OK

Cancel



3. Define Link Performance Metrics (Performance SLA)

Purpose: This is the core intelligence. You define the acceptable quality targets (latency, jitter, loss) for the link carrying the VPN traffic. The FortiGate constantly checks both wan1 and wan2 against these targets.

The screenshot shows the FortiGate SD-WAN configuration interface. The left sidebar contains the navigation menu with 'SD-WAN' selected. The main panel displays the 'Edit Performance SLA' configuration for 'VPN_Quality_Monitor'. The configuration includes:

- Name: VPN_Quality_Monitor
- Probe mode: Active (selected), Passive, Prefer Passive
- Protocol: Ping (selected), HTTP, DNS
- Server: 1.1.1.2
- Participants: All SD-WAN Members (selected), Specify
- SLA Target: ☒
- Latency threshold: 100 ms
- Jitter threshold: 30 ms
- Packet Loss threshold: 1 %
- Link Status: ☒
- Check interval: 500 ms
- Failures before inactive: 5
- Restore link after: 5 check(s)
- Actions when Inactive: Update static route (selected)

On the right, the 'SLA Details' table shows performance metrics for WAN (port4) and wan 2 (port3):

	Packet Loss	Latency	Jitter
WAN (port4)	0.00%	1.66ms	1.63ms
wan 2 (port3)	0.00%	1.66ms	1.63ms

Additional information includes links for API Preview, Edit in CLI, Performance SLA Setup Guides, Link Monitoring, SLA Targets, Documentation, Online Help, and Video Tutorials.

4. Create the SD-WAN Rule (Traffic Steering Logic)

Purpose: This rule specifically directs the VPN traffic to use the **best performing** link based on the real-time data collected

The screenshot shows the FortiGate SD-WAN configuration interface with the 'SD-WAN Rules' tab selected. The table displays the following rules:

ID	Name	Source	Destination	Criteria	Members	Hit Count
1	Optimize_Alex_VPN	NET 10	NET 30	Latency	WAN (port4), wan 2 (port3)	0
Implicit	sd-wan	all	all	Source IP	any	

The bottom of the interface shows the FortiGate logo and version 7.0.6, along with a status bar indicating 'Updated: 09:25:10'.

Cairo

DashboardNetworkSD-WANStatic RoutesPolicy RoutesRIPOSPFBGPRouting ObjectsMulticastPolicy & ObjectsSecurity ProfilesVPNUser & AuthenticationSystemSecurity FabricLog & Report

FortiGate v7.0.6

FortiGate time is out of sync.

Priority Rule

NameOptimize_Alex_VPN

Source

Source addressNET 10

User group

Destination

AddressNET 30

Protocol numberTCPUDPANYSpecify0

Internet Service

Application

Outgoing Interfaces

Select a strategy for how outgoing interfaces will be chosen.

Manual

Manually assign outgoing interfaces.

Best Quality

The interface with the best measured performance is selected.

Lowest Cost (SLA)

ID1

Last used2025/11/28 14:11:40

Hit count0

Additional Information

API PreviewEdit in CLI

SD-WAN Rules Setup Guides

Implicit RuleBest QualityLowest Cost (SLA)Maximize Bandwidth (SLA)

Documentation

Online HelpVideo Tutorials

OKCancel

Cairo

DashboardNetworkSD-WANStatic RoutesPolicy RoutesRIPOSPFBGPRouting ObjectsMulticastPolicy & ObjectsSecurity ProfilesVPNUser & AuthenticationSystemSecurity FabricLog & Report

FortiGate v7.0.6

FortiGate time is out of sync.

Priority Rule

Outgoing Interfaces

Select a strategy for how outgoing interfaces will be chosen.

Manual

Manually assign outgoing interfaces.

Best Quality

The interface with the best measured performance is selected.

Lowest Cost (SLA)

The interface that meets SLA targets is selected. When there is a tie, the interface with the lowest assigned cost is selected.

Maximize Bandwidth (SLA)

Traffic is load balanced among interfaces that meet SLA targets.

Interface preference

WAN (port4)wan 2 (port3)

Zone preference

Measured SLAVPN_Quality_Monitor

Quality criteriaLatency

Forward DSCPReverse DSCP

StatusEnableDisable

ID1

Last used2025/11/28 14:11:40

Hit count0

Additional Information

API PreviewEdit in CLI

SD-WAN Rules Setup Guides

Implicit RuleBest QualityLowest Cost (SLA)Maximize Bandwidth (SLA)

Documentation

Online HelpVideo Tutorials

5. Update the Static Route

Purpose: To make the SD-WAN rule effective, you must redirect the traffic bound for the remote VPN network away from a single physical interface and hand it over to the SD-WAN engine.

The screenshot shows the 'Edit Static Route' dialog box in the FortiGate web interface. The dialog is titled 'Edit Static Route' and has a green header bar with a warning icon and the text 'FortiGate time is out of sync.' The left sidebar shows the 'Static Routes' menu item selected. The main content area contains the following fields:

- Automatic gateway retrieval:** A toggle switch set to 'On'.
- Destination:** A dropdown menu showing 'Subnet' and 'Internet Service' with the value '30.30.30.0/255.255.255.0'.
- Interface:** A dropdown menu showing 'virtual-wan-link'.
- Comments:** A text input field with the placeholder 'Write a comment...'.
- Status:** A toggle switch set to 'Enabled'.

On the right side of the dialog, there is an 'Additional Information' section with links for 'API Preview', 'Edit in CLI', 'Documentation', 'Online Help', and 'Video Tutorials'. At the bottom of the dialog are 'OK' and 'Cancel' buttons.

The screenshot shows the 'Static Routes' table in the FortiGate web interface. The table has a green header bar with a warning icon and the text 'FortiGate time is out of sync.' The left sidebar shows the 'Static Routes' menu item selected. The table has the following columns: Destination, Gateway IP, Interface, Status, and Comments. The table contains one row of data:

Destination	Gateway IP	Interface	Status	Comments
30.30.30.0/24		virtual-wan-link	Enabled	

Conclusion

This project demonstrated the practical implementation of secure VPN solutions using FortiGate while exploring the strengths and use cases of different VPN technologies in modern network environments. Beginning with foundational concepts, the SSL VPN configuration provided secure remote access, illustrating how organizations can extend internal resources to mobile or off-site users in a controlled and authenticated manner. The IPsec VPN phase expanded on these capabilities by establishing encrypted site-to-site communication, enabling two independent networks to operate securely as a unified infrastructure. This configuration proved essential for branch connectivity, inter-office data transfer, and enterprise scalability.

The final stage integrated SD-WAN with VPN configurations to optimize performance and link utilization. By monitoring WAN health and intelligently routing VPN traffic, FortiGate's SD-WAN improved overall reliability, reduced packet loss, and ensured continuity in the presence of degraded or failing links. This demonstrated how VPN security, when combined with dynamic routing, creates a highly resilient communication environment suitable for real-world applications.

Through the implementation, testing, and performance evaluation of these solutions, the project highlighted the flexibility and robustness of Fortinet technologies. It also showed the importance of aligning VPN deployments with operational requirements—whether user access, inter-branch connectivity, or WAN performance. Overall, the project successfully achieved its objectives and provided valuable hands-on experience in designing and managing secure enterprise networks using FortiGate.